UEQManager: A Non-intrusive and Real-time System for Recognizing and Managing UEQ in Multilingual Voice Assistants

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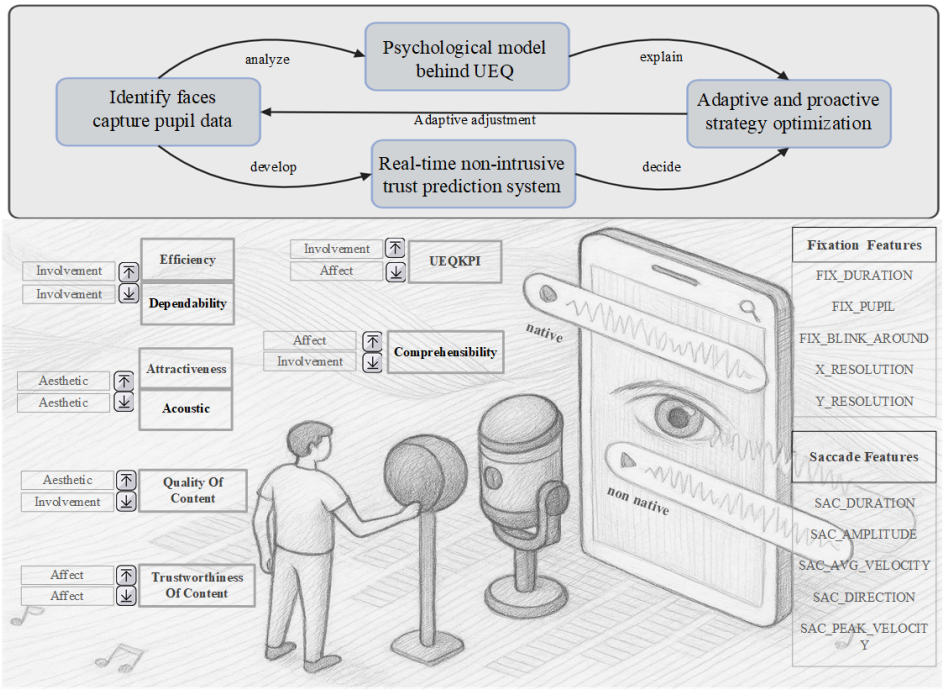


Figure 1: User experience in the interaction with multilingual VA systems

Abstract

User experience quality (UEQ) plays a critical role in multilingual intelligent voice assistant (VA), where real-time feedback directly affect interaction experience. However, current UEQ evaluation methods primarily depend on surveys, posing limitations for dynamic and user-centered adaptation. We propose a non-intrusive real-time system, UEQManager, for automated UEQ recognition

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and management. First, UEQManager incorporate interpretable deep learning models to predict UEQ in seven subdimensions. Second, leveraging LLM and expert focus group, we design adaptive interaction interface based on recognized UEQ states. Third, we implement integration system capable of real-time, non-intrusive UEQ recognition and management. We validate the effectiveness of UEQManager through a user testing experiment. Results reveal that UEQManager significantly outperformed baseline, yielding an average UEQ improvement of 27.29% over baseline and demonstrating statistical significance across all subdimensions. This work contributes a proof of concept HCI system that translates webcam gaze cues into adaptive design decisions for multilingual VAs and illustrates how designers can couple interpretable sensing with proactive interaction design.

CCS Concepts

• **Human-centered computing** → **Interaction design**; *Human computer interaction (HCI)*; • **Computer systems organization** → *Embedded systems*; Real-time systems.

Keywords

Voice Assistant, User Experience Quality, Automated Recognition, Adaptive Interface, Real-time System, Human-Centered AI

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1 Introduction

User Experience Quality (UEQ) is a crucial determinant of success in multilingual voice assistant (VA) interactions [18, 94, 102, 135, 137]. Modern VAs are now ubiquitous, with billions of devices in use worldwide [13, 39, 165]. As VA systems handle tasks from simple queries to complex dialogues, UEQ encompassing including multiple subdimensions such as attractiveness, efficiency and so on, directly impacts adoption and continued use [47, 157]. Researchers and practitioners increasingly recognize that beyond mere task completion, a positive UEQ (e.g., users finding the assistant attractive, trustworthy and so on) is vital for engagement and loyalty [30, 113, 122]. Indeed, specialized instruments like the UEQ of VA have been proposed to measure UEQ in voice interface, underscoring the importance of UEQ as a multidimensional construct (including factors such as Attractiveness) in VA design [79]. As intelligent VAs are frequently employed in multilingual environments, multilingual capability also remains crucial for achieving inclusive and effective voice interaction experiences [8, 27, 74].

However, current methods for assessing UEQ of VAs heavily rely on post-hoc questionnaires, which have significant limitations [7, 99, 106, 132]. Traditional UEQ assessments (e.g., User Experience Quality Questionnaire) capture user perceptions via Likert-scale items after an interaction session [79]. While such questionnaires provide valuable summative insights, they are retrospective and intrusive [7, 41, 61]. Users must reflect on their experience and fill out forms, often only after finishing tasks [41, 84]. This approach fails to capture moment-to-moment fluctuations in experience and can be subject to memory bias. Besides, the interruption of natural use for surveys or interviews can itself degrade the user experience [7, 21]. These limitations highlight that self-reports are insufficient for comprehensive UEQ assessment in real world contexts.

There is thus a growing need for automated and non-intrusive system to recognize UEQ during VA use. An automated UEQ recognition system would infer UEQ from behavioral and interaction cues in real time, without relying on explicit questioning [50, 99]. Prior work in intelligent interfaces and conversational agents suggests that implicit signals (e.g., hesitation in voice, repetition of queries, or body language) can indicate user states [30, 49, 99, 112, 113]. Leveraging such signals via deep learning allows continuous monitoring of UEQ [19, 30]. Unlike one-time surveys, automated recognition can provide fine-grained data on when and why the user experience falters [30]. This capability is especially important for VA interactions, where traditional GUI-based analytics (clicks, navigation patterns) are absent [131, 133]. Among various behavioral signals, eye gaze stands out as a particularly rich source of information for inferring user states [11, 49, 73]. Pupil dynamics have been shown to correlate with cognitive workload and attention, which likely map to UEQ facets [81, 139]. Therefore, this research propose to use gaze-related features as input to deep learning models for automated UEQ recognition.

To make automated UEQ recognition practically deployable in multilingual VAs, systems must be both psychologically interpretable and capable of real-time, non-intrusive sensing [9, 29, 44, 50, 85, 145]. Black-box predictions of "user is dissatisfied" are problematic for designers and stakeholders, so inferences should be explainable in human-understandable terms and grounded in psychology [2, 91]. Interpreting why specific gaze patterns signal UEQ can illuminate user behavior theories [86], motivating models that transparently link gaze features to UEQ constructs [140]. Such interpretable models both build trust—by letting developers verify that predictions make sense—and act as computational tests of cognitive theory (e.g., whether visual attention correlates with content engagement) [6, 29]. Equally, real-time processing is essential for detecting issues (confusion, dissatisfaction) as they arise and enabling on-the-fly adaptation [50]. A practical, non-intrusive pathway is webcam-based eye tracking: recent advances show common webcams can infer gaze reliably in real time without special hardware or calibration [23, 93, 143]. By leveraging camera-only interfaces, systems avoid additional sensors (e.g., wearable EEG), keep interactions natural, and unobtrusively monitor UEQ during ordinary use—aligning with ubiquitous computing trends and fitting in-home or mobile deployments [85, 145]. In sum, psychologically interpretable modeling of gaze combined with unobtrusive, webcam-based real-time sensing offers a practical and trustworthy foundation for UEQ recognition that designers can understand and that systems can act on immediately.

Ultimately, recognizing UEQ is further valuable if the system can adaptively and proactively use that information to improve the user's experience [13, 35, 105, 148, 155]. A key motivation for our work is to enable VAs not just to react to explicit commands, but to proactively adjust system behavior based on tailored UEQ

[13, 146]. If the assistant infers that the user is confused (low Comprehensibility) or unhappy with the content (low Attractiveness), it could, for example, volunteer a clarification, switch to a different strategy, or offer help—without the user having to ask [173]. Such proactive adaptation can transform a VA from a passive tool into an active partner that personalizes the interaction in real time [67]. Proactive systems underpinned by user-specific models can tailor their responses to individual preferences and contexts, creating a more user-centric dialogue flow [13]. Prior studies have shown that users respond positively when system behavior aligns with their personal style or needs [63, 115, 160]. This suggests that personalized, adaptive interface has strong potential to improve interaction quality. However, to our knowledge, a UEQ-aware adaptive interface for VA, explicitly leveraging inferred UEQ to drive real-time adaptations, has not yet been discussed, systematically studied, or implemented. Motivated by these challenges, we pose the following research questions (RQs):

RQ1: How can gaze-related features be used to recognize UEQ and its subdimensions (Attractiveness, Comprehensibility, Dependability, Efficiency, Quality of Content, Trustworthiness of Content, Acoustics)?

RQ2: What interpretable psychological models explain the associations between gaze features and UEQ subdimensions?

RQ3: Once UEQ dimensions are recognized, how can a personalized, proactive interface be designed to improve UEQ?

RQ4: How can a real-time, non-intrusive system be developed to integrating UEQ recognition and management during ongoing VA interactions?

We thus propose a proof of concept system named UEQManager to address these questions. The system consists of three technical modules that build on each other. Module 1 investigates the use of gaze features for UEQ recognition and prediction in realistic multilingual VA interactions and addresses RQ1 and RQ2. Module 1 also employ interpretability analysis to correlate users' gaze patterns with UEQ ratings, trains predictive models, and analyzes feature importance to provide psychological explanations for each UEQ subdimension. Module 2 uses the predicted UEQ values to design UEQ aware adaptive interaction strategies for multilingual voice assistants and addresses RQ3. Module 2 combines LLM based simulator with expert review to propose and refine strategies that respond to specific UEQ issues. Module 3 integrates webcam based gaze sensing, the prediction models, and the adaptive strategies into a working real time system and addresses RQ4. Finally, we run a preregistered within subjects user testing experiment with N = 18 participants. Participants interact with three simulated VAs that share the same tasks and content but differ in their adaptation logic, including a baseline system, a random strategy system, and UEQManager. Questionnaire results show that UEQManager significantly improves overall UEQ and all subdimensions compared to the two baselines.

This work makes the following contributions. Firstly, we present UEQManager, a proof-of-concept, non-intrusive real-time system that integrates webcam-based gaze sensing, interpretable UEQ prediction, and UEQ-aware adaptive interaction for multilingual voice assistants [50, 90, 99]. The main contribution focus on transferring gaze cues into adaptive design decisions. This work detail integration of full stack interface software, demonstrating feasibility for deployment. Secondly, we develop and benchmark interpretable deep learning models that use gaze features to provide moderate but robust, cross-validated predictions of overall UEQ and its seven subdimensions. We also use SHAP-based analyses to expose psychologically meaningful relationships between gaze patterns and UEQ states [9, 40, 76]. Thirdly, we propose a UEQ-aware adaptive interface design pipeline that combines LLM-based simulation with expert review to derive concrete micro-interventions for specific UEQ deficits. We instantiate this pipeline in a working proactive VA interface. Finally, we report a preregistered within-subjects user study (N = 18) that provides initial behavioral and self-report evidence of the benefits and limitations of UEQManager compared to baseline and random-strategy VAs, and we reflect on the ethical, privacy, and consent implications of webcam-based gaze tracking for real-world deployment. This work highlights a practical pathway to more personalized, empathetic voice interfaces.

2 Related Work

2.1 UEQ in Multilingual VA

User Experience Quality (UEQ) has been proved as a crucial feature in VAs as it directly influences user adoption, trust, and interaction quality [18, 94, 135]. Firstly, core components of UEQ including usability and so on, significantly affects users' willingness to adopt VAs. Studies have shown that factors such as ease of use and system efficiency are decisive in this regard [39, 116]. Secondly, prior experiences with VAs play a pivotal role in shaping current interaction quality and preferences, with positive experiences fostering trust and continued use [43]. Conversely, failures in UEQ dimensions can erode user trust. For instance, Baughan et al. examined how different types of VA failures influence trust through aspects like ability, benevolence, and integrity [10]. These studies underscore the multifaceted nature of UEQ and its critical role in the design and evaluation of VA systems.

Multilingual capability is equally essential to ensure inclusive and effective VA experiences [8, 27, 74]. Choi et al. underscore the limitations of current conversational agents in supporting users' natural communication styles and diverse cultural-linguistic contexts [27]. Wu et al. provide further insight by comparing native and non-native English speakers' interactions with VAs, revealing differences in speech adaptation strategies, interaction management, and platform preferences across smartphones and smart speakers [168]. Wenzel et al. extend this perspective by identifying culturally specific downstream harms in multilingual VA interactions, emphasizing the need for culturally aware system design [167]. Collectively, these studies affirm the importance of multilingual support in VAs for enhancing user inclusivity and improving overall interaction quality.

While existing studies have extensively explored the UEQ components and multilingual capabilities, few works have focused on how to dynamically and non-intrusively assess UEQ in multilingual interactions, especially in ways that VAs can adapt in real time. To fill this gap, this research proposes the development of a real-time, non-intrusive, and adaptive UEQ recognition interface tailored for multilingual VAs. Building upon the findings of prior work on user trust, interaction adaptation, and cross-linguistic usability challenges, this system aims to enhance both the inclusivity

and responsiveness of VA experiences through context-aware and user-centered design.

2.2 UEQ Measurements

Measuring UEQ objectively has become an active research area as HCI moves beyond solely subjective metrics [103, 113, 129, 150]. Traditionally, UEQ was measured via standardized questionnaires that assess multiple dimensions (such as attractiveness, efficiency, dependability) after interaction [75, 107, 112]. For example, Mahmood et al. measured user experience using the Questionnaire to evaluate interaction quality [102]. Klein et al. provides UEQ measurements by introducing and validating three new voice-specific UEQ scales—Response Behaviour, Response Quality, and Comprehensibility—within the modular UEQ+ framework [79]. Nevertheless, the practical application of questionnaire-based evaluations faces several limitations. Firstly, self-reporting can disrupt users' ongoing activities, thereby diverting their attention and making it unsuitable for real-time assessment [7]. Secondly, such evaluations may not reliably represent genuine trust-related behaviors, as they are susceptible to various biases, including those stemming from social desirability [106]. Therefore, recent work seeks to automate such UEQ assessments by analyzing user behavior data during the interaction [112, 113]. Furthermore, prior research has begun examining hybrid approaches that combine multiple data modalities or model types to improve UEQ recognition. For instance, some studies pair eye-tracking with other physiological signals (like heart rate or skin conductance) to capture both attentional and emotional aspects of UEQ [112]. These studies underscore the feasibility of inferring UEQ from multimodal signals (gaze, voice, etc.) in real time, moving UEQ evaluation from surveys to sensors.

Building upon the current body of research, several critical issues requires deeper investigation. Firstly, many existing studies rely on complex physiological tools, such as functional near-infrared spectroscopy, to evaluate UEQ [17, 69]. Although these approaches yield valuable insights, their practical deployment is limited due to their intrusive nature and operational complexity [120]. This highlights the necessity of exploring more user-friendly, non-intrusive alternatives [20, 169]. Secondly, a significant proportion of predictive modeling efforts in this domain tend to function as "black box" systems, offering minimal interpretability regarding the psychological processes associated with UEQ [50, 99]. Moreover, these studies often fail to consider how the relevance of specific features evolves throughout an interaction. Capturing these temporal shifts is essential for understanding the dynamic nature of UEQ, which is vital for designing VAs that can adapt in real-time to changing user states [3, 161]. Thirdly, while UEQ is widely acknowledged as a multidimensional construct, encompassing facets such as Attractiveness, Comprehensibility, and more within the context of VAs [18, 172]. The automated inference of these distinct dimensions remains largely underexplored. In response to these gaps, this research proposes a deep learning-based framework that utilizes eye-tracking data to predict both the UEQKPI and its seven sub-dimensions across VA scenarios. These dimensions include Attractiveness, Comprehensibility, Dependability, Efficiency, Quality of Content, Trustworthiness of Content, and Acoustics. By incorporating interpretable models that capture the temporal progression of eye movement features, this research aims to shed light on the cognitive and affective processes that drive UEQ in multilingual VA interactions.

2.3 Real-time and Non-intrusive Recognition System

Real-time and non-intrusive user state recognition systems have emerged as a critical development in VAs, driven by the demand for seamless, adaptive, and context-sensitive interfaces. Real-time system refers to an interface that dynamically responds to users' states through immediate processing of neurophysiological data to enhance interaction efficiency and personalization [145]. Non-intrusive system enables interaction without drawing users' primary attention, minimizing distraction, and often operating implicitly through context-aware or background processes [85]. Modern computer vision techniques enable everyday devices (like standard webcams) to perform eye-tracking and face tracking without dedicated hardware, thus "democratizing" gaze-based interaction [151, 162, 163]. Gao and Kasneci presents an interpretable and automated system for predicting user experience quality in VR locomotion by training machine learning models on eye-tracking features and using SHAP to reveal feature-level contributions to experience classification [50]. Ma er al. proposes an automated user experience quality recognition system by analyzing acoustic and social speech features to classify user satisfaction levels, offering an objective alternative to traditional self-reported UEQ measures [99]. Lee et al. developed GazePointAR, a real-time and non-intrusive multimodal VA for wearable augmented reality, which leverages eye gaze to disambiguate pronoun-based queries by integrating real-time computer vision and LLM [90]. Yen et al. introduces StoryChat, a real-time and non-intrusive viewer participation system for live streaming that uses dynamic visual narratives to proactively moderate chatrooms [170].

Collectively, these developments demonstrate a strong trend toward intelligent systems capable of real-time, non-intrusive recognition of user states. Such systems significantly advance personalized interaction by dynamically adjusting responses according to nuanced user feedback, paving the way for more intuitive, efficient, and satisfying user experiences. Building upon existing research, this study aims to develop a real-time, unobtrusive system for the automatic recognition of UEQ dimensions. The system is designed to automatically identify multiple dimensions of UEQ, thereby providing a tool to support personalized and proactive VAs and optimize user experience with VAs.

2.4 Adaptive and Proactive Interface in Multilingual VA

Recent HCI research suggests that users desire more tailored and context-aware interactions with AI systems [148]. Personalization has thus become a key factor for sustaining user engagement and satisfaction in modern AI-driven interfaces [63, 115, 160]. Zargham et al. investigates users' preferences for an adaptive VA that supports extensive customization and personalization—including voice, visual embodiment, mood and interest adaptation, and role-based access control [171]. One approach to realize adaptive and proactive VAs is to leverage AIs' generative capabilities to dynamically

adjust system responses based on the user's profile, context, or affective state [13, 35, 105, 155]. For instance, Park et al. introduced a conversation-based tutoring system that uses an LLM alongside student modeling to provide tailored instruction and guidance to each learner [118]. Oh et al. developed a proactive VA system that engages in multi-turn verbal communication to support user activities in a smart home while respecting and adapting to user agency [114]. Lee et al. presents a proactive VA system that disambiguates user queries in real time by leveraging multimodal contextual cues such as gaze, gesture, and conversation history to infer user intent without requiring fully explicit commands [90].

These developments underscore the potential of generative AI to enable adaptive, user-centered interactions in multilingual VAs. Therefore, based on the existing works, this research employs large language model agents and focus group interviews to design adaptive and proactive interaction strategies grounded in recognized UEQ within multilingual VAs. This approach aims to promote adaptive and proactive VAs, thereby facilitating its broader adoption.

3 Research Framework

This study develops a real-time, non-intrusive, interpretable UEQ recognition and management system, UEQManager for intelligent VA systems, as illustrated in the conceptual Figure 2. Specifically, this research employ deep learning model to predict UEQ (Module 1, RQ1) with interpretability analysis (Module 1, RQ2), implement adaptive interface (Module 2, RQ3), develop non-intrusive and real-time recognition system (Module 3, RQ4) and validate our proposed system UEQManager through user testing experiment (Study4). The framework specifically provides empirically grounded adaptive interface to enhance human-AI communicative efficacy across linguistic contexts.

4 UEQManager

4.1 Design Goals and System Overview

UEQManager implements the research framework introduced in Section 3 as a working system for multilingual voice assistants. Figure 3 summarizes the main components and their interactions. During an ongoing voice interaction, UEQManager continuously senses the user's eye movements, infers multidimensional UEQ, and triggers adaptive strategies that adjust the assistant's behavior. The system operates in real time and remains non intrusive, so users can interact with the assistant as usual while UEQ is monitored in the background. The system is organized into three modules that correspond to the main technical goals. Module 1 provides a gaze based prediction model that maps webcam based eye movement features to UEQ and its subdimensions in native and non native language contexts. Module 2 implements a UEQ aware adaptive interaction interface that turns predicted UEQ levels into concrete strategy choices using a library of design interventions and LLM based agents. Module 3 integrates these components into a prototype system that runs alongside a multilingual voice assistant, manages communication between the client and server, and enforces privacy preserving processing.

4.2 Module 1: Gaze-based UEQ Prediction Model

Module 1 develops a gaze based model that predicts UEQ and its subdimensions. The goal is not to reach perfect accuracy but to demonstrate that eye movement features can potentially support automated UEQ sensing in realistic multilingual VA use. This module addresses RQ1 and RQ2 by linking gaze patterns and pupil dynamics with subjective UEQ scores in native and non native language contexts. In the following, we first describe the data collection and label construction. We then present the deep learning architectures and training pipeline. Finally, we report model performance and interpretability analysis that provide psychological insights into how gaze relates to UEQ.

4.2.1 Data & Label Construction. This module employed an eye-tracking experiment conducted in native and non-native language interaction scenarios to simulate the process users undergo when interacting with multilingual VAs. We recruited a total of 60 participants (30 native and 30 non-native speakers) to obtain the dataset used in this study. The experimental procedure followed the intelligent voice interaction design of Gao et al. [48]. Each participant completed multiple interaction tasks in simulated VA scenarios. Each task consisted of three phases: (1) Situation awareness phase. Participants were introduced to a specific task-related context. This phase ensured that participants clearly understood the interaction scenario before engaging with the VA. (2) Interaction phase. Participants initiated queries by asking questions to the VA. They then listened to audio responses generated by the VA and engaged in comprehension activities. (3) UEQ Evaluation phase. Immediately after each interaction, participants rated their UEQ using UEQplus [130], providing scores across multiple dimensions with several items that served as ground-truth for regression labels. During the interaction, pupil data were recorded using the EyeLink 1000 Plus system with high temporal and spatial precision. The dataset includes fixation- and saccade-related features (detailed in Table 1). UEQ scores serve as output labels corresponding to eight core UEQ metrics across native and non-native contexts (detailed in Table 2).

A multi-stage preprocessing framework was implemented to construct standardized eye movement time-series datasets. Missing data were addressed through a dual imputation strategy: numerical features were zero-imputed, while categorical features received default identifiers. Temporal alignment was achieved by truncating sequences at the 95th percentile thresholds, with zero-padding or truncation applied to standardize sequence lengths. Independent Min-Max normalization was subsequently applied to saccadic and fixation features [1]. Multi-scale temporal enhancements were applied to the saccade and fixation features by incorporating sliding-window statistics, differential features, and their combinations to construct enriched time-series representations [153]. The finalized dataset includes normalized saccadic matrices, fixation matrices, and target UEQ metrics.

4.2.2 Model Architectures and Training Pipeline. This section presents three deep learning architectures for predicting UEQ metrics from eye-tracking data: BiLSTMAttentionNet (hereafter BiLSTM-A), CNN-BiLSTMPoolFusion (Hybrid-P), and ResCNN-BiLSTM-Attention (Hybrid-A), with their structural configurations detailed in Table 3. All our proposed models are built upon BiLSTM, given effectiveness

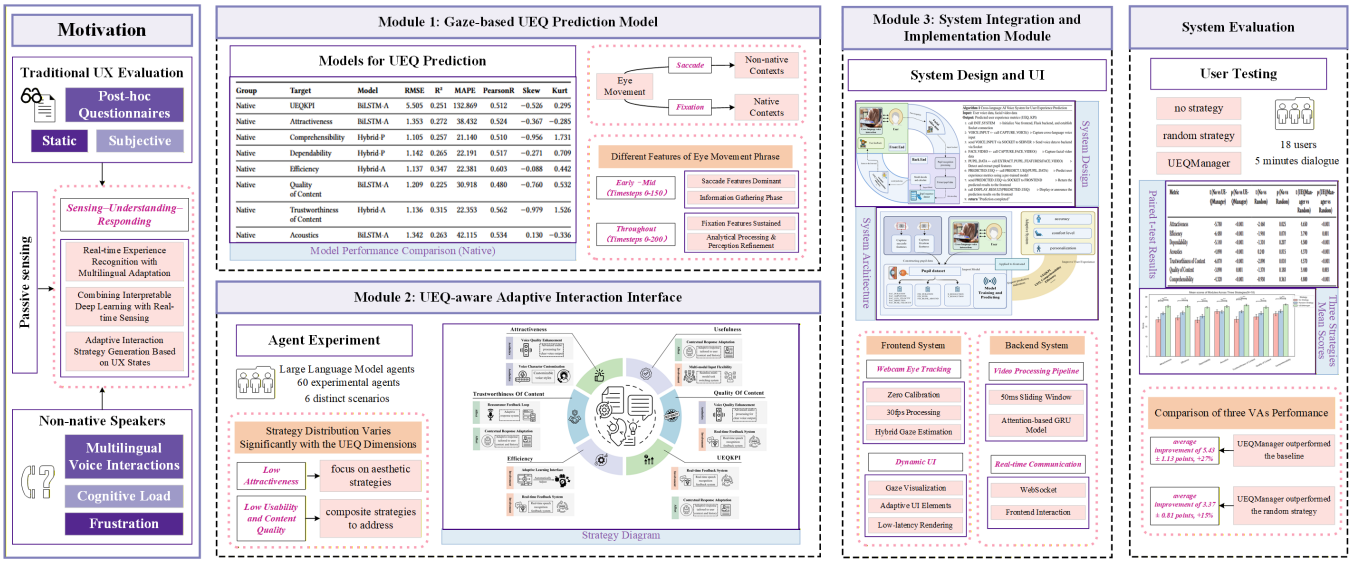


Figure 2: Research Framework

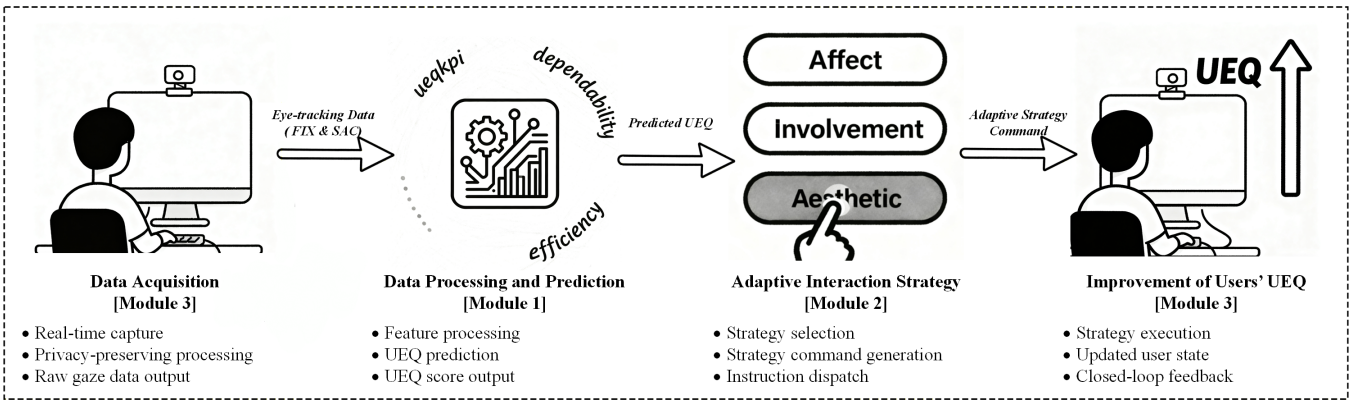


Figure 3: System Overview

in prior gaze-related sequential model [34]. The experiment conducted 5-fold cross-validation on preprocessed eye-tracking data, optimized with the AdamW optimizer and Huber loss for robust regression **[144].** Data split are by samples (not participants) to avoid individual differences. For each fold, model parameters corresponding to the highest validation R^2 were saved, while training loss and validation metrics were recorded to assess the performance of different model architectures and select the optimal model. Baseline comparisons were performed using linear regression, CNN, and a Transformer-based model. The Transformer model employed a lightweight temporal multimodal architecture with separate SAC and FIX encoders, positional encoding, a stacked Transformer encoder, cross-modal attention for feature fusion, and a regression head for prediction. This design effectively captures feature correlations and long-range dependencies within sequences. **[158].** Comprehensive model evaluation was then conducted using multiple performance metrics, including Root Mean Squared Error (RMSE) and Coefficient of Determination (R^2), enabling systematic comparison of predictive efficacy across architectures. In addition, we calculated Cohen's d for the average R^2 across different language environments to quantify effect sizes, further demonstrating the superiority of our models.

4.2.3 Prediction Performance. As shown in Table 4, cross-validation results indicate that all three proposed models consistently outperform baseline approaches (Linear, CNN, and Transformer) in terms of average R^2. For example, for the UEQKPI dimension in native language interaction scenarios, the proposed models achieve average R^2 values between 0.13 and 0.16, whereas baseline models frequently produce near-zero or negative R^2, underscoring the comparatively predictive capacity of the proposed architectures. Results further demonstrate the potential of the proposed models in capturing intrinsic relationships between eye-movement patterns and UEQ dimensions across linguistic contexts. As summarized in Appendix Table 7, the models attain best R^2 values in the range of approximately 0.19–0.35 and maintain consistently low RMSE

Table 1: Fixation and Saccade Features

Dimension	Feature	Description	Ref
Fixation Features	Current fix duration	Total duration in 10 milliseconds during which the eye remains relatively stationary. Longer durations indicate deeper visual information processing.	[80]
	Current fix pupil	Pupil diameter during fixation. Changes in pupil size reflect cognitive load, emotional state, or visual attention.	[58]
	Current fix blink around	Blink events occurring around the fixation period. Blink patterns assess attention, cognitive workload, and potential distractions or fatigue.	[56]
	Current fix x resolution	The horizontal pixel-per-degree resolution for the current fixation.	–
	Current fix y resolution	The vertical pixel-per-degree resolution for the current fixation.	–
Saccade Features	Current sac duration	Time interval in 10 milliseconds during which a saccadic eye movement occurs. Longer durations suggest more complex movements.	[164]
	Current sac amplitude	Angular distance in degrees between the start and end points of a saccade. This reflects the extent of eye movement.	[156]
	Current sac avg velocity	Average velocity in degrees per second (°/s) during a saccade. Represents the speed of eye movement.	[37]
	Current sac direction	Angular direction of the saccadic movement, indicating the spatial direction of the gaze shift.	[166]
	Current sac peak velocity	Maximum velocity reached during a saccade. Indicates the intensity or forcefulness of the movement.	[127]

Table 2: User Experience Quality Indicators and Reference Framework. All indicators are based on the framework from [16, 78, 88].

Indicator	Meaning
UEQKPI	User's overall key performance indicator
Attractiveness	Overall impression concerning the product. Do users like or dislike it?
Comprehensibility	Impression that the VA correctly understands the users' instructions and questions using natural language
Dependability	Impression to be in control of the interaction with the product
Efficiency	Impression that tasks can be finished without unnecessary effort
Quality_of_Content	Impression that the information provided by the product is up-to-date and well-prepared
Trustworthiness_of_Content	Impression that the information provided by the product is of good quality and reliable
Acoustics	Subjective experience concerning the sounds or operating noise of the product

values, reflecting stable absolute prediction errors across tasks. Hybrid-A and Hybrid-P show notable strengths in predicting Efficiency, Comprehensibility, and Trustworthiness, while BiLSTM-A performs particularly well for UEQKPI and Quality of Content. These findings highlight the promising role of eye-tracking data in supporting user experience prediction and optimization, providing robust supplementation to traditional survey methodologies [12].

For real-world deployment, predictive models are ideally evaluated with participant-level validation. The test set then represents new users who were never seen during training. We therefore additionally report a participant-level validation, with full results in Appendix Table 8. When participant-aware validation is applied, performance commonly drops because random sample splits can inadvertently exploit participant-specific patterns [33, 117]. Our results follow this pattern. For UEQKPI, the prediction decreases from 0.251 to 0.153 in the native condition. this drop mainly comes from two sources. First, our training data were collected from 60 participants, which is a reasonable scale for many CHI studies, but it is still small for learning stable, user independent models under strict participant holdout. Second, prior work across human centered sensing and time series biometrics consistently shows that moving from participant dependent splits to participant independent validation reduces reported accuracy [33, 117]. Importantly, this accuracy gap does not undermine the main goal of our work.

Table 3: Model Architectures and Characteristics

Model Type	Saccade & Fixation Branches	Final Processing	Advantages	References
BiLSTM-A	Linear layers with adaptive pooling and light 1D conv enhancement	BiLSTM + attention, LayerNorm + GELU + dropout regression head	Enhances temporal feature extraction and relevance weighting	[62, 159]
Hybrid-P	Dual 1D conv branches with BatchNorm, GELU, and max pooling	BiLSTM + fusion, dimension reduction with LayerNorm and dropout	Captures both spatial and temporal features, effectively combining CNN's local feature extraction with BiLSTM's sequence modeling	[55, 89]
Hybrid-A	Residual 1D convs with skip connections and adaptive pooling	BiLSTM + self-attention, LayerNorm + GELU + dropout regression head	Combines residual CNNs with BiLSTM and attention, improving performance by enabling better feature propagation and temporal context weighting	[57, 59]

Table 4: Cross-Validation Average R-Squared Results

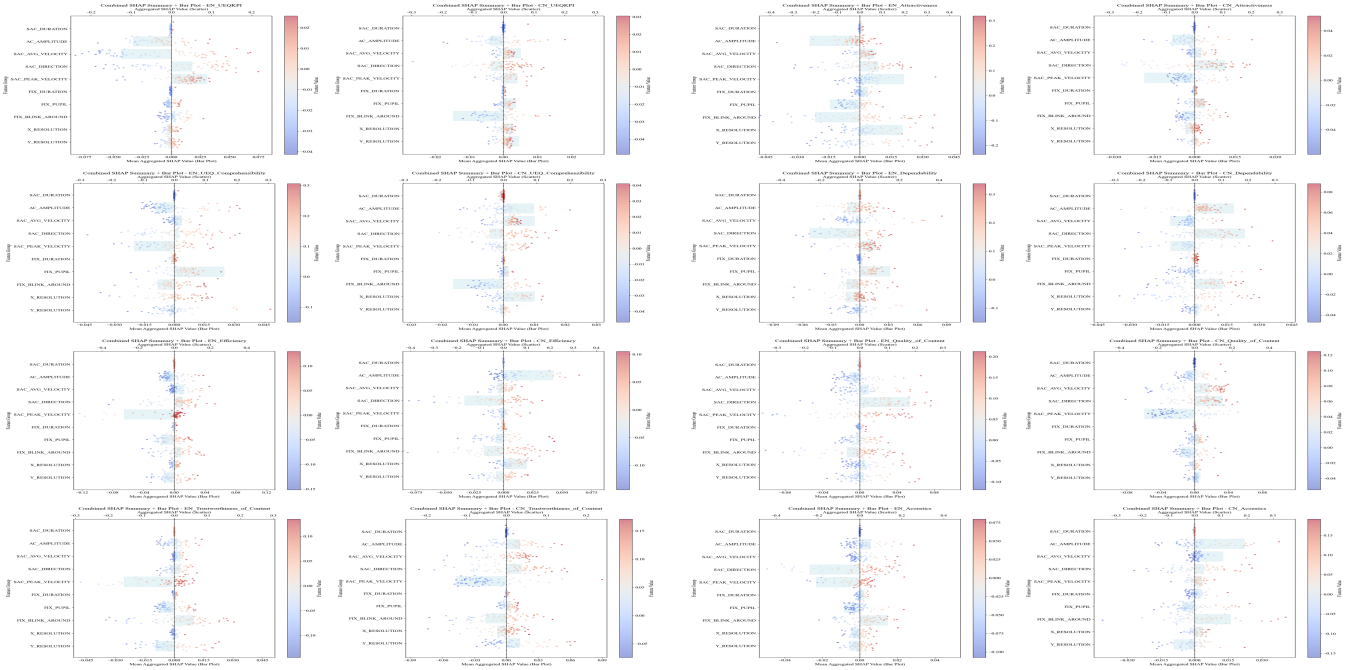
Group	Target	Hybrid-P	BiLSTM-A	Hybrid-A	Linear (baseline)	CNN (baseline)	Transformer (baseline)
Native	UEQKPI	**0.161**	0.157	0.129	-0.050	0.078	0.037
Native	Attractiveness	0.171	**0.192**	0.123	0.050	0.090	0.061
Native	Comprehensibility	0.150	0.148	**0.154**	-0.099	0.073	0.014
Native	Dependability	0.186	**0.194**	0.183	-0.099	0.054	0.021
Native	Efficiency	0.209	0.225	**0.234**	0.078	0.133	0.114
Native	Quality of Content	0.133	**0.159**	0.121	-0.007	0.082	0.036
Native	Trustworthiness of Content	**0.155**	0.108	0.148	-0.036	0.042	0.031
Native	Acoustics	0.126	**0.128**	0.101	0.045	0.093	0.070
Nonnative	UEQKPI	**0.191**	0.122	0.117	0.006	0.052	0.008
Nonnative	Attractiveness	0.103	**0.118**	0.115	0.017	0.044	0.043
Nonnative	Comprehensibility	0.244	0.191	**0.245**	0.064	0.122	0.064
Nonnative	Dependability	0.188	0.154	**0.211**	0.049	0.087	0.044
Nonnative	Efficiency	**0.219**	0.155	0.157	0.075	0.134	0.084
Nonnative	Quality of Content	**0.210**	0.197	0.155	0.067	0.098	0.033
Nonnative	Trustworthiness of Content	**0.202**	0.161	0.171	0.061	0.131	0.094
Nonnative	Acoustics	0.111	**0.120**	0.112	-0.008	0.028	0.048

The predictive module is designed to support a proof of concept system for gaze based UEQ prediction that enables adaptive voice assistants, not to maximize predictive accuracy in isolation. In practical deployments, strict zero shot generalization is also not the only option, since a short onboarding step or lightweight per user calibration is often feasible in gaze based systems. For example, current work has demonstrated calibration procedures that reduce user burden [121]. Vision research has shown that a model can be adapted to a new person with only a handful of calibration samples [119]. Finally, we expect participant level generalization to improve with larger and more diverse datasets, as has been repeatedly observed in gaze modeling when scaling participant counts and data volume [82]

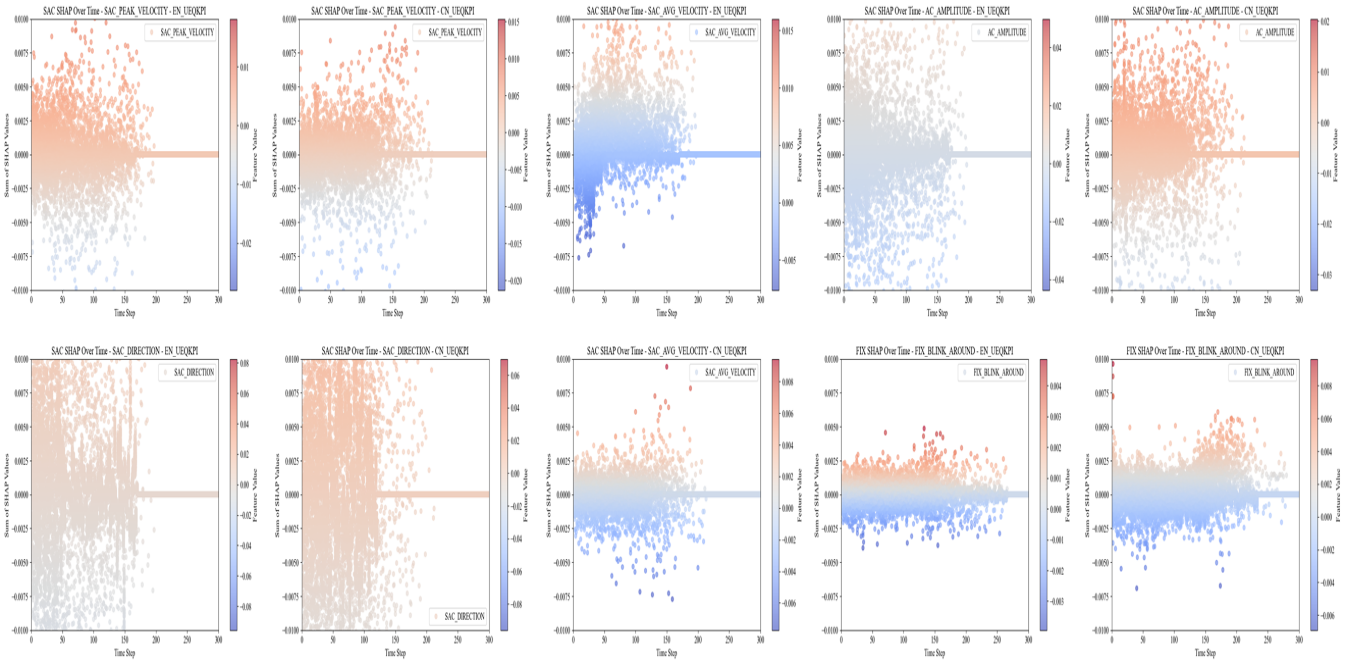
4.2.4 Interpretability and Psychological Insights. SHAP [96, 97, 154] quantifies the contribution of each feature to model outputs. In this section, we first select the best-performing model for each prediction target as the subject of explanation. The top ten samples from the test set are chosen as the background data, and their temporal lengths are aligned to a unified number of time steps. Interpretability analysis results are shown in Table 3, providing correlational evidence and insights into the relative importance of gaze features for cross-linguistic UEQ prediction. Figure 4a displays the SHAP summary and bar plots for eye movement features, where bar lengths correspond to mean SHAP values indicating feature contributions to User Experience Questionnaire (UEQ) KPI prediction. Figure 4b illustrates the temporal distribution of SHAP values across time steps, with darker hues representing higher feature values. Our analysis of the UEQKPI reveals:

First, the interpretability analysis identifies language-dependent correlations between pupillary dynamics and UEQ metrics. The non-native language context UEQKPI demonstrates stronger associations with saccadic activities, particularly smaller saccadic amplitudes and lower average saccadic velocities. Conversely, the native language context UEQKPI correlates more closely with fixation patterns, specifically reduced blink frequency during fixations. This dichotomy suggests divergent cognitive processing strategies: non-native language processing induces heightened cognitive load [83], prompting users to adopt chunked information processing with reduced perceptual intake per fixation to enhance comprehension [125]. This phenomenon aligns with the "trade-off" theory, where saccadic efficiency is reduced to maintain processing accuracy under cognitive stress **[71]**. In native language contexts, suppressed blink frequency during fixations [101] indicates intensive engagement with complex content, which paradoxically reduces environmental familiarity and increases processing costs, ultimately diminishing user experience.

Second, temporal analysis reveals phase-dependent contributions: saccadic features predominantly influence early-mid phase predictions (0-150 time steps), whereas fixation features maintain predictive power across all phases (0-200 time steps) as shown in Figure 4b. This pattern reflects the dual cognitive demands in voice-assisted interactions – concurrent information gathering and processing during early phases, followed by sustained analytical processing during later fixations, which are cognitively linked to mental representations [87]. This pattern is consistent with the foundational cognitive model that early saccades guide attention,



(a) SHAP feature importance ranking with eye movement metrics



(b) Temporal SHAP value distribution across interaction phases

Figure 4: Interpretability analysis of the proposed model: (a) demonstrates the global feature importance through mean absolute SHAP values, (b) visualizes the temporal evolution of SHAP contributions during native/non-native interactions.

while subsequent fixations integrate information into working memory, directly influencing user experience perception **[123]**. The findings elucidate the cognitive mechanics of user experience formation in voice-AI interactions: users establish baseline experience through saccadic-fixation cycles, then refine perceptions through later-stage fixations. Notably, acoustic quality predictions exhibit language-specific patterns: larger saccadic amplitudes positively correlate with acoustic ratings in native language contexts, while smaller amplitudes show stronger associations in non-native language contexts. This suggests that native language proficiency enables efficient acoustic information processing through expanded saccadic scanning, whereas non-native processing imposes cognitive constraints [83], forcing users into restricted scanning patterns that degrade perceived audio quality, particularly for less proficient users. This observation also aligns with models of eye movement control, which propose that reduced processing efficiency due to high cognitive load dynamically adjusts attention by limiting saccade length **[126]**.

4.3 Module 2: UEQ-aware Adaptive Interaction Interface

Module 2 translates predicted UEQ into concrete adaptive interaction strategies, shifting from sensing user states (Module 1) to deciding how the system should respond when a UEQ issue is detected. Addressing RQ3, UEQManager implements a lightweight rule-based interface that inspects the current predicted score of each UEQ dimension and, when any score falls below a configurable threshold, selects a dimension-specific micro-intervention pattern aligned with the implied deficit. For example, low Comprehensibility triggers clarification and step-by-step guidance, whereas low Trustworthiness of Content triggers explanation and reassurance. Large language model (LLM) agents then instantiate these patterns as context-appropriate utterances conditioned on the current task and dialogue history. We position this LLM-based adaptation as a scalable complement to expert intervention strategies rather than a full replacement: recent HCI work has used generative models as design partners and as low-cost proxies for user and expert feedback when exhaustive expert involvement is impractical [92, 95, 118]. Following this line, we use LLM agents to explore a large space of candidate micro-interventions under diverse UEQ profiles, then have experts review, filter, and refine them in focus groups to ensure the final strategies are professional, feasible, and suitable for multilingual VA deployment, while also covering edge cases that expert rules may miss.

4.3.1 Strategy Space and Adaptive Agent Design. Building on previous prediction module, adaptive module aims to optimize interaction strategies based on predicted UEQ levels. The study employed 60 experimental agents using the DeepSeek-R1 model [32], aiming to enhance user experience through adaptive responses. Specifically, these 60 agents were configured as simulators of user preference, where each agent was assigned a distinct, pre-defined UEQ level to model a diverse range of real user satisfaction. Experiments were divided into six scenarios corresponding to UEQ dimensions. Real-time prompts were generated for each scenario, requiring LLM agents to recommend interaction strategies based on users' trust levels. Specifically, agents received users' UEQ scores and selected optimal strategies for improvement. In this process, agents simulated the action of selecting the optimal strategy from a provided strategy library that would best address the needs implied by that specific UEQ level, thereby facilitating the discovery of personalized adaptive policies. The strategies focused on three dimensions: Involvement, Aesthetics, and Affect [152]. The design and application of information systems in cross-language intelligent voice interaction have garnered significant attention [109], making personalized strategies a critical design element. The study analyzed the dimensional distribution of LLM-selected strategies across scenarios and extracted frequently implemented measures. Figure 5 illustrates the agent experiment design process.

4.3.2 Agent-based Simulation and Strategy Analysis. Through analysis of 60 scenarios across 8 UEQ dimensions, Figure 6(a) demonstrates the proportional distribution of strategy dimensions across conditions, and Figure 6(b) identifies the two most frequent intervention measures per condition. The findings provide theoretical and practical foundations for designing cross-linguistic voice systems. Key results reveal:

First, when Acoustics and Attractiveness scores are low, Aesthetics dominates strategy selection (78.9% and 76.1%), emphasizing sensory-level experiences. Primary interventions include Voice Quality Enhancement and Voice Character Customization. Specifically, cross-linguistic voice systems should prioritize auditory aesthetics optimization for first-time users through advanced sound enhancement technologies (e.g., noise reduction algorithms) and diversified voice persona customization. This aligns with established findings that aesthetic design critically impacts initial user impressions and emotional engagement in novel interfaces [25].

Second, when Comprehension levels are low, the system tends to enhance users' perception of usefulness through the Affect dimension(83.3%). This approach primarily involves Contextual Response Adaptation. Specifically, when users exhibit skepticism regarding content usefulness, the system dynamically adjusts responses to better align with users' immediate context and interaction patterns, consistent with previous findings that increased engagement strengthens perceived usefulness [38]. Conversely, when predicting low Quality of Content evaluations, the strategy predominantly focuses on Aesthetics (50.0%), aligning with established research demonstrating that improved content clarity enhances quality perception [147].

Third, significant variations in strategy distribution across UEQ dimensions reflect diverse user needs in cross-linguistic interactions. As Figure 6(a) illustrates, Dependability and Efficiency deficits prioritize Involvement strategies (61.7% and 89.4%) through Adaptive Learning Interfaces and Real-time Feedback Systems. For instance, systems dynamically optimize interface layouts during international travel scenarios, prioritizing high-frequency modules like translation/emergency assistance while providing instantaneous progress updates. In contrast, Trustworthiness of Content deficiencies emphasize Affect strategies (71.1%), requiring emotional reassurance through empathetic voice tones and trust-building dialogues. Systemic UEQ KPI declines necessitate balanced dimensional allocations, highlighting the need for context-aware strategy optimization that harmonizes technical solutions with human factors [5, 51].

Algorithm 2 Speech UEQ Analysis System

```
procedure SpeechUEQAnalysis
    // Phase 1: Data Collection
    Initialize folders and data structures
    PreventSystemSleep()
    ueqDimensions ← ["Attractiveness", "Efficiency", "Usefulness", "TrustworthinessOfContent",
"QualityOfContent", "UEQKPI"]
    allResponses ← [ ]
    for each dimension in ueqDimensions do
        scenarioResponses ← [ ]
        for each batch in totalBatches do
            batchResponses ← [ ]
            for each call in batchCalls do
                // Generate prompt and call API
                dimensionDescription ← GetDimensionDescription(dimension)
                opennessList ← RandomlySelectOpenness()
                prompt ← FormatPrompt(dimension, dimensionDescription, opennessList)
                response ← CallAPIWithRetry(prompt) // API call with retry mechanism
                // Store response
                if response is valid then
                    status ← "success"
                else
                    status ← "failure"
                end if
                responseData ← CreateResponseObject(call, dimension, response, status)
                batchResponses.Add(responseData)
                scenarioResponses.Add(responseData)
                allResponses.Add(responseData)
                WaitBetweenCalls()
            end for
            SaveBatchData(batchResponses, dimension, batch)
            WaitBetweenBatches()
        end for
        SaveScenarioSummary(scenarioResponses, dimension)
        WaitBetweenScenarios()
    end for
    SaveAllResponses(allResponses)
    AllowSystemSleep()
    // Phase 2: Data Analysis
    Load response data
    Initialize analysis structures
    for each dimension in ueqDimensions do
        Filter successful responses
        Extract and count dimensions and measures
        Calculate statistics
        Store analysis results
    end for
    Save extraction details
    // Phase 3: Visualization
    if results exist then
        Setup visualization environment
        Create dimension distribution visualizations
        Create measure frequency visualizations
        Create comparative visualizations
        Save visualization outputs
    end if
end procedure
```

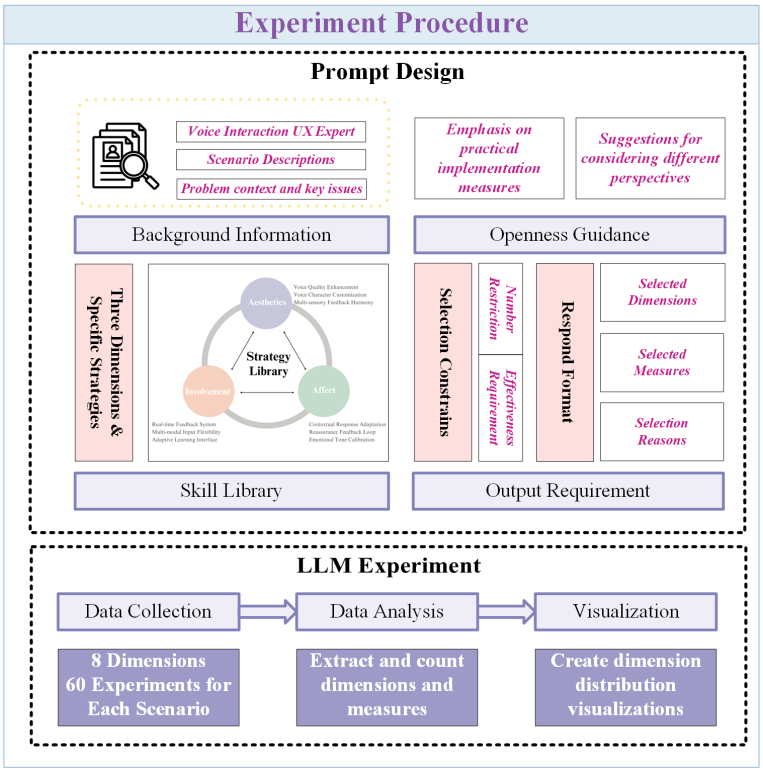
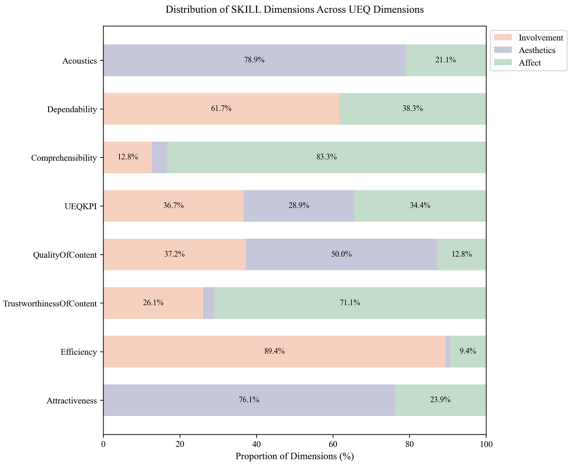
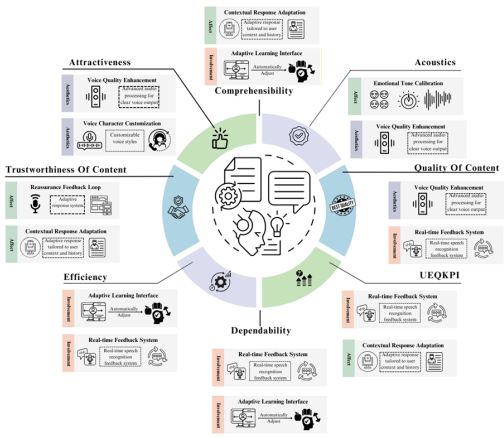


Figure 5: Adaptive Agent Design



(a)Strategy Distribution Across Scenarios



(b)Strategy Diagram

Figure 6: Strategy Distribution and Diagram

4.3.3 Expert Review and Refinement. To evaluate and refine the professionalism of LLM agents' strategy selection, focus group interviews were conducted to reveal diverse perspectives through group interactions [68]. Focus group interview design followed Liu et al. [95] and Choi et al. [26]. During the discussions, host maintained a neutral stance, with open-ended questions to guide conversation, and ensured equal opportunities for all participants to speak. Participants were recruited via purposive sampling to ensure diverse backgrounds. Interviews centered on the adaptive interaction interface, rationality of scenario-specific adjustments, impact of dynamic strategies on user experience, and practical implications. Semi-structured interviews included open-ended questions,

followed by a 7-point Likert scale evaluation (1="completely unprofessional" to 7="highly professional") to quantify methodological credibility [124].

To validate LLM-generated strategies, we conducted expert evaluations via focus group interviews with 8 HCI specialists using Likert 7-point scales. Results indicate moderate professional adequacy (Mean=5.59, SD=1.50) for LLM suggestions. For users with lower attractiveness scores, experts recommend enhancing cultural identification through dialect-friendly approaches and emotional connections (e.g., incorporating brand storytelling and local dialect greetings). In terms of usefulness, improving correction accuracy by implementing multi-turn dialogue mechanisms that allow users to modify instructions at any time, coupled with instant confirmation feedback. Regarding efficiency, streamlining interaction processes through intuitive interface design to reduce operational time consumption, alongside dynamic resource preloading to minimize waiting periods. For trustworthiness, increasing transparency in interaction logic and reinforcing user confidence via emotional reassurance designs (e.g., mild-toned language and encouraging feedback), while maintaining consistent punctuation and structural integrity as specified.Integrating expert insights with LLM strategies enables precise balance between technical capabilities and humanistic needs.

4.4 Module 3: System Integration and Implementation

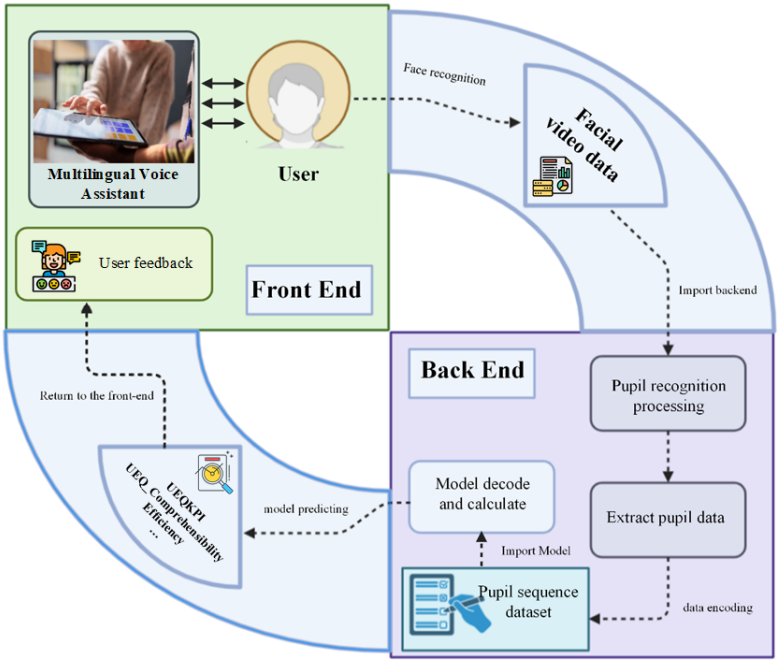
Module 3 integrates the prediction and adaptation components into a working proof of concept system that we call UEQManager. This module addresses RQ4 by demonstrating that UEQ can be recognized and managed in real time without intrusive sensing hardware. It connects webcam based gaze tracking on the client side with UEQ prediction models and the adaptive strategy module on the server side. In this section, we first describe the overall system design and user interface. We then detail the back end architecture that links gaze processing, UEQ prediction, and proactive responses. We also discuss implementation choices, runtime performance, and privacy considerations for camera based sensing.

4.4.1 System Design and User Interface. We propose a prototype of real-time UEQ prediction and adaption system (UEQManager) to enhance the ability of the Multilingual VA to accurately assess and manage user UEQ levels. As shown in Figure 7, given the advantages of the front-end-back-end separation architecture in terms of lightweight design and system stability [53], this study adopted Vue 3 for front-end development and Flask for back-end implementation. The front-end employs a web camera-based eye tracker [24] to continuously capture pupil movement data, which is subsequently transmitted to the back-end through socket network communication. The back-end system integrates the prediction module and UEQ-aware module to perform analytical computations and adaptive adjustments.

4.4.2 System Architecture and Data Flow. UEQManager operates as a real-time non-intrusive prediction and adaption framework that continuously monitors users' UEQ without requiring task interruption or physical sensors, thereby dynamically enhancing intelligent systems' response capabilities. WebGazer is a JavaScript-based eye-tracking tool that can be directly integrated into the system frontend, utilizing the user's webcam to infer gaze points in real time, thus eliminating the need for additional hardware support. As shown in Appendix Figure 12, the system first acquires basic eye movement data (position and timestamp) using a webcam-based online eye-tracking system [65]. The system employs facial and ocular detection algorithms to identify pupil positions, represents the eyes as image patches, and maps ocular image features to screen coordinates via ridge regression, thereby providing real-time gaze point coordinates (x, y) and pupil detection results for each frame, which are used to calculate eye movement metrics such as pupil diameter. To address the impact of pupil constriction caused by intense illumination on measurement accuracy, this study normalizes the collected eye movement features based on pupil size changes induced by lighting conditions [104], ensuring standardized measurements across different lighting environments. Subsequently, we employ an online eye-movement classification model with temporal convolutional networks to process the acquired basic eye movement data [42], classifying fixations and saccades and computing the required eye movement metrics. Among these, the "Current fix blink around" metric is calculated using a pre-trained Vision Transformer (ViT) model [134]. Finally, for different UEQ metrics, we use the best-performing prediction model to predict UEQ and further leverage the varying performance across different UEQ dimensions in adaptive interface to propose interaction strategies, thus enhancing the user experience.

For the evaluation of system performance, we evaluated the real-time performance of UEQManager by quantifying key metrics such as end-to-end latency and throughput. Following established methodologies in HCI for real-time eye-tracking systems [90], we measured latency as the time from frame capture to UEQ prediction output using high-precision timing functions (Python's `time.perf_counter()`). In a commodity laptop setup with a standard webcam, we ran 12 independent trials (60 s each) and observed an end-to-end latency of 39.2 ± 3.1 ms (mean ± SD across trials), with a 95% interval for the mean of [37.2, 41.2] ms; throughput averaged 24.8 ± 2.6 fps, with a 95% interval of [23.1, 26.5] fps. These results remain below the commonly used interactive threshold (<50 ms for perceived real-time response in gaze-based interfaces), indicating that UEQManager provides responsive feedback in real time. Throughput was assessed as the effective frames per second (fps) processed without loss, and the achieved throughput is consistent with the requirements of practical webcam-based real-time eye-tracking pipelines.

4.4.3 Ethics and Privacy Concerns. UEQManager relies on camera based sensing to recognize users' pupil dynamics and estimate UEQ. This design raises clear ethical and privacy concerns. Eye tracking and gaze data can function as biometric information and can reveal sensitive personal traits. Prior work in HCI has shown that gaze behaviour supports user authentication and may expose private attributes and states [31, 72, 141]. Research on camera glasses and webcams also documents strong user concern about being recorded without awareness or control [14, 36, 100]. We therefore design UEQManager around explicit informed consent and a privacy preserving processing pipeline.



Algorithm 1 Pseudocode for Real-time Non-intrusive User Experience Quality (UEQ) Prediction System

Initialization:
Initialize front-end/back-end separation architecture, activate front-end WebGazer.js eye tracker
Initialize back-end Flask service and socket network communication
Load calibration interface and calibration markers
Front-end processing flow:
Display calibration markers for user calibration
while video stream not ended **do**
  Image capture: Acquire real-time video frames using camera
  Image preprocessing:
    Convert to grayscale image
    Apply noise filtering algorithms
  Pupil detection:
    Extract eye region of interest (ROI)
    Perform local enhancement, edge detection and binarization
    Calculate pupil center coordinates (achieving sub-pixel accuracy using geometric transformation matrix)
  Record current frame timestamp and pupil center coordinates
  Data transmission: Transmit eye movement data to back-end via socket
end while
Back-end processing flow:
while receiving new eye movement data **do**
  Generate eye metrics: Calculate ten eye movement metrics based on continuous video frame data
  Model prediction:
    Input eye movement metrics into following prediction models:
      (a) BiLSTMAttentionNet
      (b) CNN-BiLSTMPoolFusion
      (c) ResCNN-BiLSTM-Attention
  Integrate model outputs to predict user's UEQ level
  Return UEQ results to intelligent voice assistant for dynamically adjusting user experience
end while

Figure 7: System Design

Before any sensing starts, the system presents a consent dialog. The dialog explains what will be captured, how the data will be processed, and which parts of the system run locally. Users are informed that UEQManager only uses the front camera to extract pupil related features that are needed for UEQ recognition and adaptive responses. The raw camera frames are processed on the client device, where we detect eyes and compute pupil size and movement features. Only compact feature vectors and the predicted UEQ labels are sent to the server for logging and adaptation logic. This edge centred design avoids transmission of full video streams and reduces both network load and the risk of unintended secondary use of rich visual data.

All feature data in transit between client and server is protected with standard transport encryption and short lived session keys. The server keeps only the minimal state that is necessary to compute the current UEQ and to control the adaptation. Raw pupil features are stored in memory only during this computation stage and are then discarded. We do not build any long term user profiles from gaze data and we do not store raw images or videos. This follows the principle that sensitive biometric measurements should be processed in an immediate use and discard fashion whenever possible, similar to privacy preserving pipelines proposed for gaze data in prior work [31, 141].

These design decisions are informed by HCI research that links privacy concerns to a lack of transparency and control over sensing. Studies of webcams, wearable cameras and IoT devices show that users feel more comfortable when systems clearly signal when sensing happens and provide understandable controls [14, 36, 100, 138]. Camera based sensing is already common in voice assistants, smart displays and driver assistance systems. Prior work suggests that once users understand what is captured and why, privacy concerns become more contextual and less salient [72, 138]. In our post study interviews, participants reported that they were initially cautious about camera use. After the consent process and a short explanation of the local processing and deletion policy, most participants reported that privacy was no longer a dominant concern. Many felt that the benefits of adaptive support justified the use of the camera. These findings suggest that privacy aware design can support both ethical use of eye based sensing and practical deployment of UEQManager like systems.

4.5 Example Use Case

This process illustrates the real-time UEQ recognition and management workflow of UEQManager during an everyday cooking query with a multilingual VA (Figure 8). After the user accepts the privacy policy, the VA starts the conversation and offers an initial answer. In parallel, the system monitors eye movements with the webcam and records raw pupil positions and trajectories in the background. The eye-movement pipeline encrypts these data and extracts fixation and saccade features, which are then fed into the deep learning models. The models estimate the current scores of the UEQ dimensions and can, for example, detect that Comprehensibility is low while other dimensions remain acceptable. When such a deficit is detected, an LLM-based agent selects a concrete strategy from the design library, such as slowing the pace, adding step-by-step guidance, or rephrasing the answer in simpler terms, and passes this plan to the VA. The VA adapts its next turns accordingly, helping the user follow the instructions with less effort and feel that the assistant is attentive, useful, and responsive. Ongoing

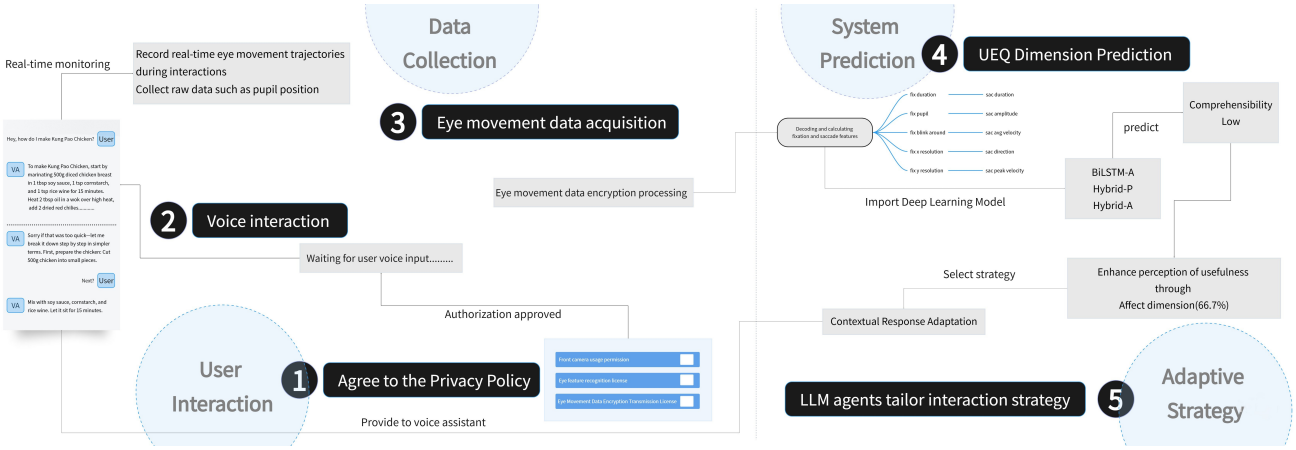


Figure 8: UserCase

real-time monitoring closes the loop, indicating whether the adaptation improves UEQ and whether further adjustments are needed, so the user's moment-to-moment state directly drives continuous refinement of the interaction.

5 User Testing

After developing UEQManager, we conducted a user testing study to evaluate the effectiveness of UEQManager as a whole system. The goal of this study is to test whether the adaptive interface improves user experience compared to non adaptive baselines. We use a within subjects design with three simulated VAs that share the similar tasks and content but differ in their adaptation logic. UEQ is measured through standardized questionnaires and behavioral speech features. This section describes the experimental procedure, analysis methods, and main findings.

5.1 Experiment Design

This research performed a preregistered user testing experiment (N=18) to validate the effectiveness of our proposed system, UEQManager. The experimental procedure is shown in Figure 9. By developing an experimental platform to simulate the interaction between users and VAs, we designed three virtual VAs: a blank group (no strategy), a random strategy group, and an UEQManager group. Each of the 18 users interacted with all three VAs, with each dialogue lasting approximately 5 minutes. The experiment design followed methodologies from prior studies [4, 45, 66, 77], with controlling for variables such as dialogue topics. Immediately after each dialogue session, participants completed the UEQ+ questionnaires to validate effectiveness of our proposed system [78]. The questionnaire scores used a 7-point scale. Each dimension of the UEQ has 4 items. By calculating the mean scores and standard deviations for each group, conducting pairwise paired t-tests, and computing the corresponding effect sizes, we evaluate the effectiveness of UEQManager in improving UEQ.

To address the limitations of questionnaire-based evaluations, audio features were extracted from video recordings of user testing sessions to evaluate speech behavior across baseline, random, and UEQManager conditions. Video paths and predefined temporal segments were specified for videos, with segments corresponding to groups: baseline, random, and UEQManager. Audio extraction was performed using FFmpeg to isolate monaural audio clips at a 16 kHz sampling rate from each segment, ensuring computational efficiency and minimizing resource usage[15]. The extracted audio data were processed to compute key speech features, including speech rate (via zero-crossing rate approximation) and jitter (short-term fundamental frequency perturbation). These features were derived using vectorized operations in NumPy and SciPy. Such acoustic measures are widely employed in voice pathology and quality assessment, providing objective indicators of fluency and vocal stability [70, 98, 99]. Statistical analysis employed non-parametric methods suitable for small sample sizes and non-normal distributions common in speech data. For overall group differences, a permutation-based repeated-measures ANOVA was implemented, computing the F-statistic from sums of squares and permuting within-subject observations across 5000 iterations to derive empirical p-values [54]. Pairwise comparisons utilized bootstrap resampling (5000 iterations) on differences, estimating p-values via the proportion of resampled means crossing zero, with effect sizes (r) derived from z-scores[149]. These approaches ensure robust inference in behavioral speech studies, accounting for inter-subject variability[28].

5.2 Results and Discussion

The mean scores, significance test results, and effect sizes are presented in Table 5 ,Table 6 and Figure 10. The results showed that the VA with UEQManager significantly outperformed the blank group (baseline system) (average improvement: 5.43±1.13 UEQ points, +27%) and the random strategy group (excluding potential effects of introducing additional random strategies) (average improvement: 3.37±0.81 UEQ points, +15%). This indicates that applying appropriate adaptive strategies when specific UEQ dimensions are low can enhance overall user experience levels. Notably, the adaptive group demonstrated the most significant improvement in Trustworthiness of Content compared to the no-strategy group (average improvement: 7.22±1.44 points).This suggests our adaptive strategy enhanced user trust in information provided by intelligent VAs.

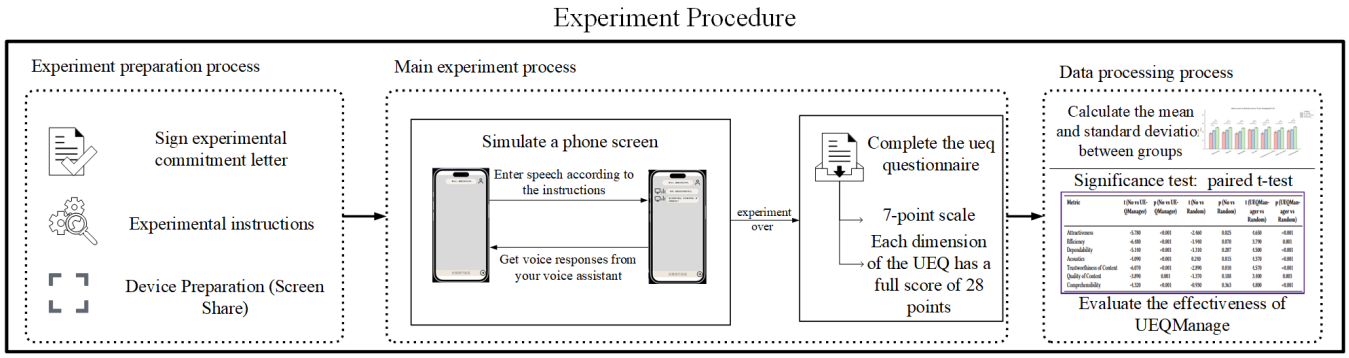


Figure 9: Experimental Procedure

Table 5: Paired t-test and Bootstrap results for interaction experience metrics

Metric	t (None vs UEQManager)	p (None vs UEQManager)	t (None vs Random)	p (None vs Random)	t (UEQManager vs Random)	p (UEQManager vs Random)
Attractiveness	-5.780	<0.001	-2.460	0.025	4.650	<0.001
Efficiency	-6.480	<0.001	-1.940	0.070	3.790	0.001
Dependability	-5.140	<0.001	-1.310	0.207	4.500	<0.001
Acoustics	-4.090	<0.001	0.240	0.815	4.370	<0.001
Trustworthiness of Content	-6.070	<0.001	-2.890	0.010	4.570	<0.001
Quality of Content	-3.890	0.001	-1.370	0.188	3.400	0.003
Comprehensibility	-4.320	<0.001	-0.930	0.363	4.800	<0.001
Speech Rate		<0.001		0.313		0.036
Jitter		0.036		0.504		0.377

Note: Speech Rate and Jitter are behavior metrics, and the rest are interaction experience metrics

Table 6: ANOVA Results and Effect Sizes

Aspect	F	p	Eta Squared	Cohen f
Attractiveness	22.584	< 0.001	0.470	0.941^{***}
Efficiency	18.927	< 0.001	0.426	0.862^{***}
Dependability	19.483	< 0.001	0.433	0.874^{***}
Acoustics	4.043	0.023	0.137	0.398^{**}
Trust	22.614	< 0.001	0.470	0.942^{***}
Quality	8.259	0.001	0.245	0.569^{***}
Comprehensibility	13.206	< 0.001	0.341	0.720^{***}
Speech Rate	4.448	0.019	0.270	0.609^{*}
Jitter	1.272	0.301	0.096	0.326

Note: Effect size for Cohen's f: * small ($f \geq 0.1$), ** medium ($f \geq 0.25$), *** large ($f \geq 0.4$). Speech Rate and Jitter are behavior metrics, and the rest are interaction experience metrics.

To validate the effectiveness of UEQManager, comparisons between the No-Strategy and UEQManager groups showed that all seven metrics yielded p-values below 0.05 with negative t-values, indicating significantly higher mean scores under UEQManager. Similarly, in comparisons between the Adaptive and Random Strategy groups, all seven metrics showed p-values below 0.05 with positive t-values, again demonstrating that UEQManager significantly outperformed the random baseline. Across the three groups, effect-size calculations further confirmed the practical importance of these improvements, with Cohen's f values consistently exceeding 0.35.

The analysis revealed significant differences in speech rate and jitter features 5 6, highlighting the impact of UEQManager interfaces on user vocal behavior during testing. Speech rate showed an overall group effect (permutation ANOVA F = 4.448, p = 0.0192), with pairwise significance between baseline and UEQManager (bootstrap p = 0.0000, r = 0.000) and random and UEQManager (bootstrap p = 0.0364, r = 0.580), indicating faster rates in UEQManager conditions, potentially reflecting reduced cognitive load and improved system alignment[28]. Jitter exhibited pairwise significance between baseline and UEQManager (bootstrap p = 0.0360, r = 0.582) and no significant differences between baseline and random (bootstrap p = 0.5040, r = 0.185), suggesting lower vocal perturbations in UEQManager scenarios, which may correlate with decreased anxiety or enhanced stability[46]. These findings support the efficacy of UEQManager systems in enhancing speech naturalness and user experience, consistent with prior voice quality research.

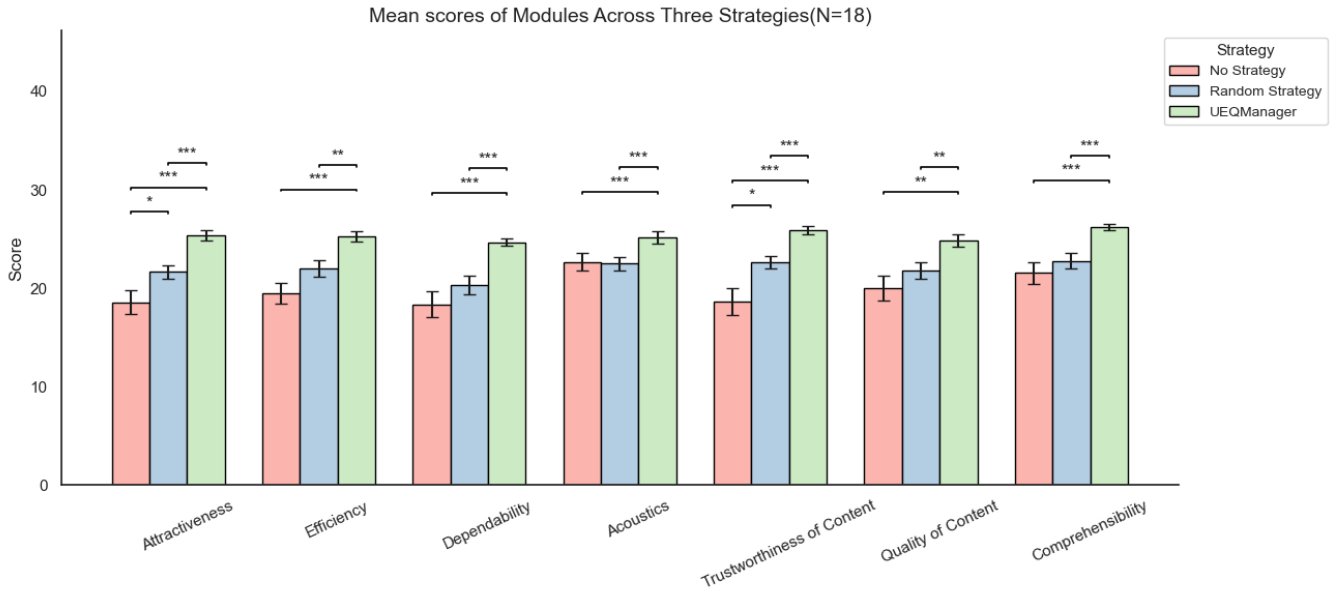


Figure 10: Mean scores of Three strategies

6 General Discussion & Conclusion

This work presents UEQManager as a proof of concept HCI system that links webcam gaze sensing, interpretable UEQ prediction, and adaptive interface design for multilingual VAs. The core contribution is a pipeline in which gaze cues are translated into UEQ predictions and then into concrete micro level design decisions during ongoing interactions. For example, when the system collects pupil data in real time and predicts that the Quality of Content index is low, it adopts a feedback strategy that surfaces fresher results and explains data sources, which in turn improves perceived content quality. This study addresses four critical challenges in eye movement tracking-based UEQ prediction for intelligent VAs: (1) leveraging eye movement features to identify users' UEQ Key Performance Indicators (KPI) and its seven sub-dimensions, (2) uncovering the psychological models underlying UEQ metrics, (3) constructing a real-time non-intrusive UEQ monitoring system, and (4) developing adaptive strategies with dynamic prediction adjustments based on recognized UEQ states. Our findings are illustrated in Figure 11. For example, when the system collects user pupil data in real time and predicts that the Quality of Content index is low, it will adopt a real-time feedback strategy to provide users with the latest results and explain the update time and data source, and indicate that users will be informed of any updates to improve users' satisfaction with the Quality of Content.

Taken together, our results provide a feasibility proof-of-concept system transferring eye cues into adaptive interaction decisions. First, across both native and non-native interactions, our gaze-based deep learning models achieve moderate but robust predictions of overall UEQ and all seven subdimensions, consistently outperforming linear, CNN, and Transformer baselines. This suggests that webcam eye-movement features are a viable signal for automated UEQ sensing in realistic multilingual VA use. Second, SHAP-based interpretability analyses reveal systematic, language-dependent patterns. These correlations provide psychologically grounded insight into how cognitive load and engagement shape UEQ. The results also point to which gaze cues designers can attend to when reasoning about user experience. Third, by integrating these models into UEQManager and coupling them with a UEQ-aware adaptive interface, we build a real-time, non-intrusive proof-of-concept system that turns recognized UEQ deficits into concrete micro-interventions. A preregistered within-subjects study (N = 18) shows consistent UEQ improvements over baseline and random-strategy VAs, together with changes in speech rate and jitter that suggest more fluent, confident interaction. We view these findings as initial but encouraging evidence that UEQ-aware adaptation can improve the moment-to-moment experience of multilingual voice assistants.

6.1 Theoretical Implication

Through addressing specific research questions, several broader theoretical implications emerge from this work. Firstly, this work contributes to the theoretical understanding of multimodal UEQ measurement in user interface software and technology [103, 113, 129, 150]. This work provide empirical evidence that objective behavioral signals can serve as proxies for subjective user experience dimensions. This finding supports an expanded view of UEQ theory where user behavior is an integral part of the UEQ construct [92, 113]. Traditionally, UEQ has been conceptualized through self-reported perceptions and post-use reflections [79]. This work suggests that user experience can also be understood as a continuous, observable process, manifested through physiological and behavioral cues during interaction. Interpretability analysis bridges the gap between external behavior and internal experience [29, 44]. For instance, theories of engagement often describe psychological

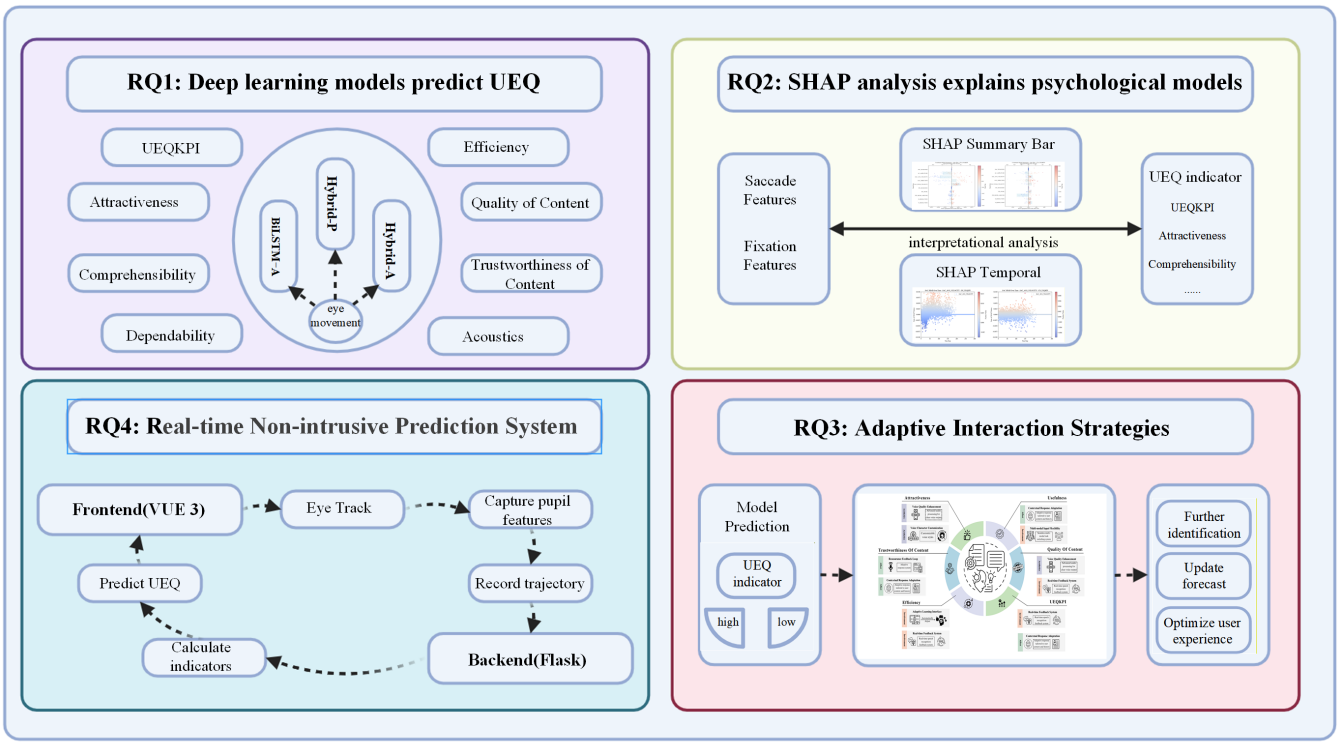


Figure 11: Research results summary

immersion, and gaze-based measures offer a way to quantify that immersion by observing attention focus [110]. This integration of behavioral data into UEQ models could lead to richer theoretical frameworks that account for dynamic user-state changes, not just static impressions. Furthermore, this work was validated across native and nonnative users, which implies a potential universality in gaze-UEQ relationships. The core cognitive and affective responses we tracked appear to transcend linguistic context, lending weight to the theory that certain aspects of user experience are common across diverse language proficiency. This cross-cultural consistency is an important theoretical insight for global UEQ research [8, 27, 74].

Secondly, by linking specific gaze patterns to particular UEQ subdimensions, this work expands theoretical model of user attention and cognition in voice interactions [22, 108]. This work findings indicate that users exhibit specific gaze behavior patterns consistent with changes in UEQ. These observations align with and extend existing psychological theories, including cognitive load theory and information processing theory [22, 64, 111]. This work effectively map those theoretical concepts onto the domain of VAs. Even in primarily auditory interfaces, the visual channel (gaze) plays a role in how users process and react to information [60]. This highlights a theoretical notion that VA interactions are not purely auditory, as inherently multimodal experiences when a user's environment and body language are considered [102]. Thus, comprehensive theory of voice-based UEQ should account for how users' multi-sensory participates in the interaction loop for further expansion. This work provides a concrete step in that direction, suggesting which gaze metrics correspond to which underlying mental state during a conversation with an AI assistant.

Thirdly, this proactive adaptation mechanism has theoretical implications for interaction design and agency in AI systems as human-AI loop [52, 136]. In HCI theory, there has long been a careful balance between system initiative and user control [3, 142]. This work findings suggest that giving the system a degree of agency to act proactively can improve the interaction, so long as the system's actions are appropriately informed by the user's state. This contributes to a theoretical model where the VA is seen not just as a reactive tool, but as a collaborative agent that shares responsibility for the dialogue outcome [13, 35, 105]. System intervenes at the right moments (when specific user experience is declining) and with the right adjustments (tailored to the detected problem), which validates theories suggesting that users will welcome proactive help if it clearly aligns with their current needs and goals [114, 173]. Additionally, this work touches on personalization theory in user experience [160]. Classic models of personalization distinguish between long-term personalization (adapting to user traits or preferences) and session-based adaptation [137]. This work shows these are complementary, as assistant performs immediate, short-term personalization (adjusting to the user's present state) which in turn likely contributes to longer-term personalization (as the system learns which strategies work best for that user). Theoretically, this hints at a multi-layered personalization framework for intelligent

systems, where instant inference of user state augments the traditional user profile to drive adaptation [137, 160]. It underscores the notion that user experience at each moment matters and can be optimized, not just the overall experience over many sessions.

6.2 Practical Implication

Beyond theory, this work yields concrete practical implications for the design of future intelligent VAs. Firstly, developers of commercial VAs (e.g., Alexa, Siri, Google Assistant) can incorporate this gaze-based UEQ recognition mechanism to make their platforms user-aware in real time. This could dramatically improve user satisfaction. For instance, a smart speaker could detect if a user is decreasing UEQ by irrelevant answers and proactively say, "Let me try that again in a different way," or provide a helpful clarification. By addressing issues immediately, the assistant can prevent annoyances from accumulating into a negative overall experience [114, 173]. This work's system is readily applicable using existing device hardware (cameras in phones, tablets, smart displays), suggesting a near-term opportunity for industry adoption.

Secondly, this system provides a method for continuous, in-situ UEQ evaluation that can benefit UEQ researchers and designers. Instead of relying solely on scheduled user testing or periodic surveys, product teams could deploy this capability to silently monitor UEQ quality for beta users or volunteer participants [103, 113, 129]. For example, during a new feature rollout for a VA, the system could flag drops in Efficiency or Attractiveness scores in real usage, alerting the design team to possible usability issues or content problems. This represents a shift towards more data-driven UEQ design for voice interfaces [113, 128]. Decisions can be informed by quantitative UEQ measures gathered in real time, complementing qualitative user feedback. Additionally, the richness of the data (multiple UEQ dimensions over time) can help pinpoint exactly where in a dialog things went wrong, enabling targeted improvements.

Thirdly, this work demonstrated a practical use of large language models to handle dynamic user adaptation, which has implications for AI system development. Rather than scripting numerous if-then rules for how an assistant should respond to various user states, developers can leverage an LLM to interpret a high-level instruction (like "the user seems confused, explain more simply") and generate an appropriate response [102]. This greatly reduces the development overhead for creating adaptive dialogues, because the model handles linguistic nuance and context[114]. The implication is that with the combination of UEQ sensing and powerful generative models, even relatively small development teams could endow a VA with sophisticated, context-aware behaviors. This practical implication shifts some design effort from writing dialogue variants to carefully engineering prompts and state-to-action mappings for the LLM. This work suggests that future VAs will likely use LLMs as a component to achieve personalization [118], allowing VAs to dynamically change personality, tone, or content level of detail in response to recognized user needs. This approach can also extend to other modalities (e.g., changing a displayed image or GUI element via generative AI) as guided by the user's state. Taken together with the gaze-based UEQ model, UEQManager ground an adaptive interface in which system behaviour is driven by explicit representations of the user's state. Rather than fully automating the assistant, designers can decide when VAs should take initiative, when it should defer to the user, and how to surface these choices, thus distributing agency between human and AI in a more accountable way.

6.3 Limitations and Directions for Future Studies

This research proposes several limitations that should be addressed in future studies. Firstly, we acknowledge the limitation of the current prediction accuracy. The primary contribution of this work is the design and demonstration of a proof-of-concept system. Future studies could focus on enhancing model robustness and predictive performance based on our benchmark. Secondly, the interpretability analysis presented offers only correlational evidence. It does not constitute a direct empirical validation of the underlying specific mechanisms between eye movements and UEQ scores, as explicitly validating these complex cognitive pathways was not the core focus of this study. Thirdly, the system's robustness under varied operational conditions (e.g., lighting, sound, background, temperature, and humidity) was not evaluated, which is crucial for practical deployability. Future studies could test the system through field experiment to enhance the ecological validity of this proof-of-concept system. Finally, despite safeguards, we acknowledge that camera-based pupil sensing may still raise residual ethical and privacy concerns. Future research should explore stronger technical and organizational safeguards, as well as participatory design with diverse stakeholders. We view user state awareness as a promising route toward more human-centered AI, and believe that addressing this challenge is preferable to abandoning such technologies altogether.

6.4 Conclusion

In conclusion, this work presents UEQManager as a proof-of-concept system for gaze-informed UEQ recognition and management in multilingual voice assistants. By grounding UEQ predictions in interpretable gaze features, we provide HCI researchers and designers with a concrete mapping between observable eye behavior and specific UEQ dimensions. This mapping functions as an explicit, gaze-based user model that can drive adaptive interface behaviour while keeping users' experiential state and agency at the centre of design. These mappings inform the design of adaptive interface. When the system detects gaze patterns associated with low scores in a given dimension, it selects targeted micro-interventions, for example clarification, pacing or content reformulation strategies. User testing shows that this pipeline from gaze cues to adaptive decisions can yield measurable improvements in users' reported UEQ and related speech behavior, even with moderate predictive accuracy. More broadly, the work illustrates how webcam-based gaze sensing can act as a practical design resource for real-time user state awareness. It also highlights the need for further work on robustness, ethics and long-term deployment in everyday VA use.

UEQManager offers a design template for how future conversational VAs infer aspects of user experience from gaze cues and adapt accordingly. By linking UEQ with adaptive system design, our work suggests a possible path toward more user-centered VAs.

Demonstrating gains in empathy, trust, and effectiveness in everyday use remains an important direction for future work. We envision a future where interacting with VAs feels more comfortable and supportive. Our results represent an early step toward exploring this direction rather than a demonstration of end-to-end human-level interaction quality.

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A Appendix: Model Performance

Table 7: Model Performance Metrics for Best Models

Group	Target	Model	RMSE	R^2	MAPE	PearsonR	Skew	Kurt
Native	UEQKPI	BiLSTM-A	5.505	0.251	132.869	0.512	−0.526	0.295
Native	Attractiveness	BiLSTM-A	1.353	0.272	38.432	0.524	−0.367	−0.285
Native	Comprehensibility	Hybrid-P	1.105	0.257	21.140	0.510	−0.956	1.731
Native	Dependability	Hybrid-P	1.142	0.265	22.191	0.517	−0.271	0.709
Native	Efficiency	Hybrid-A	1.137	0.347	22.381	0.603	−0.088	0.442
Native	Quality of Content	BiLSTM-A	1.209	0.225	30.918	0.480	−0.760	0.532
Native	Trustworthiness of Content	Hybrid-A	1.136	0.315	22.353	0.562	−0.979	1.526
Native	Acoustics	BiLSTM-A	1.342	0.263	42.115	0.534	0.130	−0.336
Nonnative	UEQKPI	Hybrid-P	6.490	0.243	140.066	0.495	−0.057	−0.535
Nonnative	Attractiveness	BiLSTM-A	1.419	0.214	43.086	0.463	−0.399	0.293
Nonnative	Comprehensibility	Hybrid-A	1.103	0.345	20.895	0.589	−1.030	1.793
Nonnative	Dependability	Hybrid-A	1.101	0.264	20.395	0.527	−0.248	0.215
Nonnative	Efficiency	Hybrid-P	1.280	0.322	29.491	0.568	−0.279	0.209
Nonnative	Quality of Content	BiLSTM-A	1.118	0.283	23.990	0.538	−0.321	0.606
Nonnative	Trustworthiness of Content	Hybrid-P	1.090	0.305	19.572	0.552	−0.292	0.642
Nonnative	Acoustics	BiLSTM-A	1.324	0.195	43.251	0.443	0.105	−0.656

Table 8: Model Performance Across Participant-Level Splits

Group	Target	Model	RMSE	R^2	MAPE	PearsonR	Skew	Kurt
Native	UEQKPI	CNN	4.824	0.153	387.824	0.404	−0.244	−0.182
Native	Attractiveness	CNN	1.488	0.093	53.648	0.389	0.124	−0.167
Native	Comprehensibility	Transformer	0.904	0.030	14.214	0.180	0.405	0.593
Native	Dependability	Linear	1.061	0.044	17.310	0.440	−0.418	1.929
Native	Efficiency	Hybrid-P	1.428	0.156	33.573	0.409	−0.053	−0.656
Native	Quality of Content	CNN	1.023	0.104	22.495	0.335	0.095	−0.304
Native	Trustworthiness of Content	Linear	1.109	0.100	16.701	0.531	−0.422	1.924
Native	Acoustics	Hybrid-A	1.160	0.069	34.770	0.335	−0.360	−0.142
Nonnative	UEQKPI	Hybrid-P	8.483	0.017	120.147	0.148	0.163	−1.022
Nonnative	Attractiveness	BiLSTM-A	1.329	0.043	37.428	0.216	0.034	0.226
Nonnative	Comprehensibility	Transformer	0.913	0.074	13.785	0.316	1.141	1.977
Nonnative	Dependability	CNN	1.266	0.137	28.531	0.370	0.773	1.288
Nonnative	Efficiency	Hybrid-A	1.358	0.094	27.329	0.371	0.265	−1.003
Nonnative	Quality of Content	CNN	1.158	0.107	25.145	0.346	0.237	−0.472
Nonnative	Trustworthiness of Content	BiLSTM-A	1.295	0.173	30.142	0.446	1.409	2.809
Nonnative	Acoustics	BiLSTM-A	1.253	0.038	42.201	0.199	−0.687	0.021

B Appendix: System Architecture

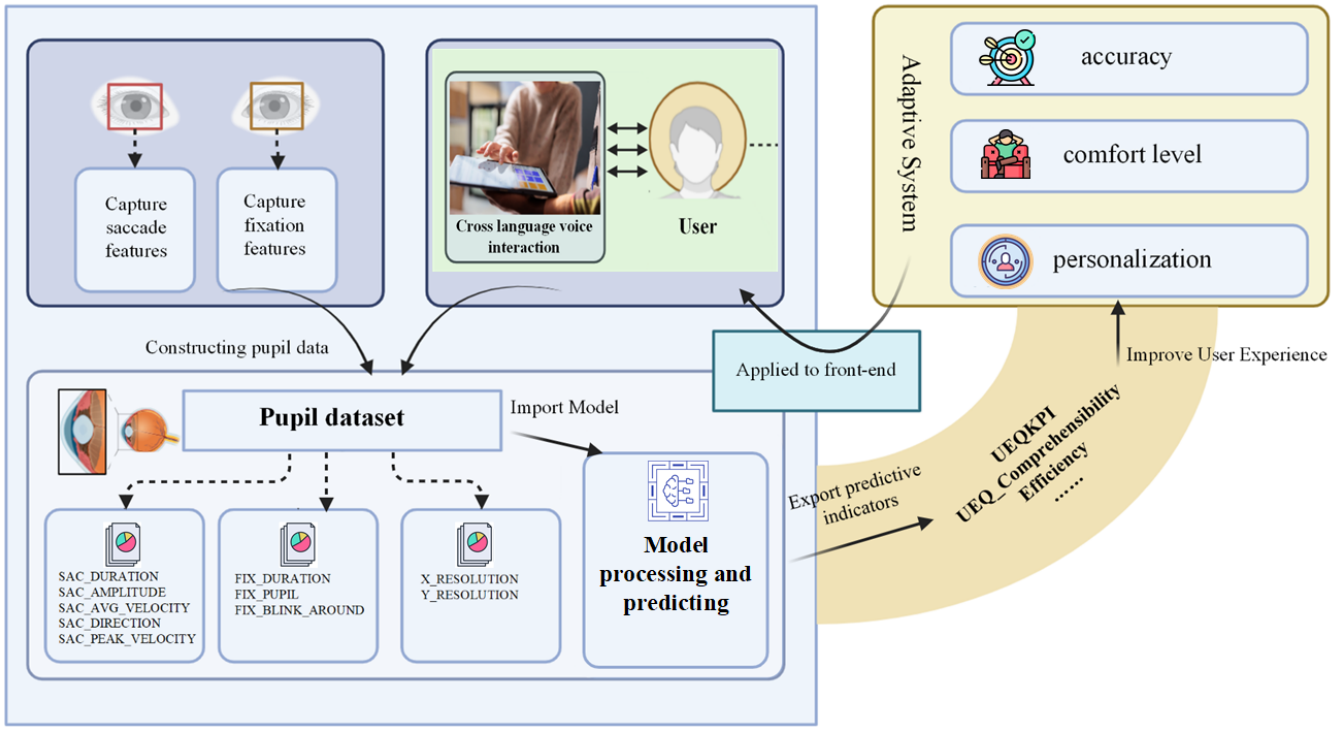


Figure 12: System Architecture

C Appendix: Participant Data

Table 9: Participant Demographic Information and Multidimensional Expert Rating Responses

ID	Gender	Age	Research Field	Degree	Attractiveness	Efficiency	Usefulness	Credibility	Quality	Acoustics	Compreh.	Depend.	UEQ-KPI	Key Opinion
1	Male	23	Voice Alert HCI	Master	5	3	4	4	5	6	4	4	5	System deficiency prioritization
2	Female	24	Edu. System Similarity	Master	6	6	6	6	6	6	7	7	6	Engagement-driven success criteria
3	Female	23	Voice Interaction	Master	3	4	5	6	5	7	4	7	6	Dimension-specific adaptation
4	Female	23	Fund Perf. Models	Master	5	3	6	4	3	-	-	-	4	Subjective perception alignment
5	Female	36	HCI & Design	Master	3	7	7	5	4	-	-	-	6	Usability-first optimization
6	Female	25	DL Architectures	Master	3	7	7	7	7	7	5	7	7	Command accuracy imperative
7	Female	24	LLM Systems	Master	7	7	7	7	7	5	7	7	7	Experience-centric optimization
8	Male	24	Supply Chain Mgmt	Master	2	7	6	7	1	6	6	7	3	Contextual dimension adaptation
9	Female	21	HCI	Bachelor	5	7	6	6	7	6	7	7	7	Imitate the way humans naturally communicate